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LIS 440

wk12: Pandas with YOUR Assigned State's Ten Years of Year-End Data

11/26/25

Code

```
# -----  
# File & Path Setup  
# -----  
  
# Import the necessary packages  
import pandas as pd  
import os  
from pathlib import Path  
  
# Set current working directory (CWD) : root folder  
root = Path(__file__).parent.parent.resolve()  
  
# Show CWD  
print("Current Working Directory: ", Path.cwd())  
  
# Set CWD: Data  
data = root / 'data'  
# Set CWD: Data > Raw  
data_raw = data / 'raw'  
# Set CWD: Data > Processed  
data_processed = root / 'data' / 'processed'  
# Set CWD: Scripts  
scripts = root / 'scripts'  
  
# Read MO Avg Home Data  
mo_home_data = data_processed / 'mo_10yr_avghomeval.csv'  
  
# Create dataframe of MO average home val data  
mo_df = pd.read_csv(mo_home_data)  
print(mo_df)  
  
# -----  
# Clean to Only Last 10 Years of Data  
# -----  
  
# Convert 'YearEndData' to datetime b/c I forgot to filter for that on excel  
mo_df['YearEndData'] = pd.to_datetime(mo_df['YearEndData'], format='%m/%d/%y')  
  
# Get the latest year from the data  
latest_year = mo_df['YearEndData'].dt.year.max()  
  
# Filter for rows within the last 10 years  
MOTenYearEndDF = mo_df[mo_df['YearEndData'].dt.year >= (latest_year - 9)]  
  
# Clean up 'AverageHomeValue' column by removing $ and commas, then convert to float  
MOTenYearEndDF['AverageHomeValue'] = (  
    MOTenYearEndDF['AverageHomeValue']  
    .replace('[\$,]', '', regex=True)
```

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        .astype(float)
    )

# Quick check
print(MOTenYearEndDF) # Keep MOTenYearEndDF per the instructions

# -----
# Data Analysis Steps for Missouri (MO)
# -----

# Subset series of home values
MOYearEndDatesSeries = MOTenYearEndDF['AverageHomeValue']

# Display the series
print("\nYear End Averages:")
print(MOYearEndDatesSeries)

# Print 3rd, 6th, and 9th values
print("\nThird year:", MOYearEndDatesSeries.iloc[2])
print("Sixth year:", MOYearEndDatesSeries.iloc[5])
print("Ninth year:", MOYearEndDatesSeries.iloc[8])

# Create custom row labels
row_labels = MOTenYearEndDF['YearEndData'].dt.strftime('MOTenYearEnd20%y')
MOTenYearEndDF.index = row_labels

# Print first, fifth, and last rows using custom labels
labels_list = MOTenYearEndDF.index.tolist()

print("\nFirst row:")
print(MOTenYearEndDF.loc[labels_list[0]])

print("\nFifth row:")
print(MOTenYearEndDF.loc[labels_list[4]])

print("\nLast row:")
print(MOTenYearEndDF.loc[labels_list[-1]])

```

Shell

```

>>> %Run mo_avghomeval.py
Current Working Directory: /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S
440/Lectures/Week 13/pandas_state_activity/scripts
  YearEndData AverageHomeValue
0  12/31/00   $102,486.25
1  12/31/01   $109,484.25
2  12/31/02   $116,499.76
3  12/31/03   $123,798.34
4  12/31/04   $129,699.80
5  12/31/05   $136,394.66
6  12/31/06   $140,794.88
7  12/31/07   $140,469.55
8  12/31/08   $135,934.51
9  12/31/09   $131,217.42

```

10	12/31/10	\$127,031.51
11	12/31/11	\$120,023.51
12	12/31/12	\$122,297.69
13	12/31/13	\$127,011.40
14	12/31/14	\$130,747.26
15	12/31/15	\$137,997.32
16	12/31/16	\$144,866.54
17	12/31/17	\$152,810.99
18	12/31/18	\$161,261.82
19	12/31/19	\$169,402.55
20	12/31/20	\$188,220.49
21	12/31/21	\$212,720.11
22	12/31/22	\$233,421.78
23	12/31/23	\$243,331.05
24	12/31/24	\$253,125.77

/Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S 440/Lectures/Week 13/pandas_state_activity/scripts/mo_avghomeval.py:46: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

MOTenYearEndDF['AverageHomeValue'] = (

	YearEndData	AverageHomeValue
15	2015-12-31	137997.32
16	2016-12-31	144866.54
17	2017-12-31	152810.99
18	2018-12-31	161261.82
19	2019-12-31	169402.55
20	2020-12-31	188220.49
21	2021-12-31	212720.11
22	2022-12-31	233421.78
23	2023-12-31	243331.05
24	2024-12-31	253125.77

Year End Averages:

15	137997.32
16	144866.54
17	152810.99
18	161261.82
19	169402.55
20	188220.49
21	212720.11
22	233421.78
23	243331.05
24	253125.77

Name: AverageHomeValue, dtype: float64

Third year: 152810.99

Sixth year: 188220.49

Ninth year: 243331.05

First row:

```
YearEndData    2015-12-31 00:00:00
AverageHomeValue    137997.32
Name: MOYearEnd2015, dtype: object
```

Fifth row:

```
YearEndData    2019-12-31 00:00:00
AverageHomeValue    169402.55
Name: MOYearEnd2019, dtype: object
```

Last row:

```
YearEndData    2024-12-31 00:00:00
AverageHomeValue    253125.77
Name: MOYearEnd2024, dtype: object
```

```
>>>
```